

The Shift Lemma and the j -formula for all hooks: structural proof by induction on the arm length

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Abstract

We isolate a clean operator identity inside the Iwahori–Hecke algebra $H_q(S_n)$ — the *Shift Lemma*

$$R'_a R'_{a+1} \cdots R'_b = (T_{a-1}+1)(T_a+1) \cdots (T_{b-1}+1) R'_{a-1} R'_a \cdots R'_{b-1}, \quad 2 \leq a \leq b \leq n$$

— and use it to give a structural proof of the j -formula for **every hook** $\lambda = (k, 1^{n-k})$ in the rank-zero regime $n \geq 2k$.

Theorem (main). For every $k \geq 2$ and every $n \geq 2k$,

$$B_{j_0(\lambda)}^{(\lambda)} V_\lambda := R'_{j_0(\lambda)} R'_{j_0(\lambda)+1} \cdots R'_n V_\lambda \subseteq E_1^-|_{V_\lambda}, \quad j_0(\lambda) = n - k + 2.$$

In particular $\Pi^{S_n}|_{V_\lambda} = 0$ unconditionally for every hook λ in the rank-zero regime.

The proof is a clean induction on k . Base cases $k = 2$ (May-11) and $k = 1$ (sign rep, trivial). Inductive step: the Phase A decomposition $E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S$ splits the image $R'_n V_\lambda$ into two pieces:

- The $\Phi_*(V_{\mu_1})$ piece, with $\mu_1 = (k-1, 1^{n-k-1})$, is handled by the May-13 r1 commutation (which carries through hooks unconditionally) combined with the inductive hypothesis at $k-1$.
- The same-row part S , isomorphic as an $H_q(S_{n-2})$ -module to $V_{\nu'}$ with $\nu' = (k-2, 1^{n-k})$, is killed by an application of the Shift Lemma: the iteration $R'_{j_0} \cdots R'_{n-1}|_S$ rearranges into T -factors $\times$ an inner iteration that, via the iso $S \cong V_{\nu'}$ and the inductive hypothesis at $k-2$, equals zero.

The Shift Lemma is the operator-theoretic engine: it lets us *shift the entire iteration down by one in indices*, at the cost of multiplying through by a row of commuting T -factors. This shift-by-one is exactly what's needed to align the same-row iteration on λ with the sharp j -formula iteration on ν' .

Consequences. Beyond the hook j -formula itself:

1. The May-12 evening-3 paper's $\dim B = 3$ for $\lambda = (4, 2, 1^{n-6})$ becomes *unconditional* (the upper bound relied on the hook j -formula at $k = 4$, which was conditional then; this paper closes that gap).
2. The same-row-dies conjecture for $\lambda = (k, k, 1^q)$ reduces to the j -formula for the sub-shape $\nu = (k, k-2, 1^q)$ via the same Shift Lemma machinery (concrete next target on this thread).
3. The framework is shape-agnostic: any same-row-dies argument can be carried out via Shift Lemma plus a j -formula on the sub-shape.

The mathematical content is short and self-contained; the heaviest input is the May-11/May-12 hook base cases and the Phase A endpoint (May-19).

1 Setup

Notation follows the May-13 r1 paper (2026-05-13-r1-inductive-reduction.tex) and the May-12 hook paper (2026-05-12-hook-j-formula.tex). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ satisfying the usual quadratic relations $(T_i - q)(T_i + 1) = 0$ and the braid relations. For $\lambda \vdash n$, V_λ is the seminormal cell module with basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

Definition 1. For $1 \leq k \leq n$, set

$$R'_k := (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1) \in H_q(S_n),$$

with the convention $R'_1 = 1$ (empty product). For $1 \leq a \leq b \leq n$, set

$$P_{a,b} := R'_a R'_{a+1} \cdots R'_b,$$

with the convention $P_{a,b} = 1$ when $a > b$. For a partition λ ,

$$B_j^{(\lambda)} V_\lambda := P_{j,n} V_\lambda = R'_j R'_{j+1} \cdots R'_n V_\lambda, \quad j_0(\lambda) := \ell(\lambda) + 1.$$

Write $E_j^\pm$ for the $\pm$ -eigenspaces of T_j on V_λ (eigenvalues q resp. -1).

2 The Shift Lemma

Lemma 2 (Shift Lemma). For $2 \leq a \leq b \leq n$, inside $H_q(S_n)$,

$$P_{a,b} = (T_{a-1} + 1)(T_a + 1) \cdots (T_{b-1} + 1) \cdot P_{a-1,b-1}. \quad (1)$$

Proof. By induction on $b - a$.

Base case $b = a$: $P_{a,a} = R'_a = (T_{a-1} + 1) R'_{a-1} = (T_{a-1} + 1) P_{a-1,a-1}$. Matches (1).

Inductive step: Assume (1) for $P_{a,b-1}$ (i.e., the lemma for length $b - 1 - a$). We have

$$P_{a,b} = P_{a,b-1} R'_b \stackrel{\text{IH}}{=} (T_{a-1} + 1)(T_a + 1) \cdots (T_{b-2} + 1) P_{a-1,b-2} R'_b.$$

Unfold $R'_b = (T_{b-1} + 1) R'_{b-1}$:

$$P_{a,b} = (T_{a-1} + 1)(T_a + 1) \cdots (T_{b-2} + 1) P_{a-1,b-2} (T_{b-1} + 1) R'_{b-1}. \quad (2)$$

Key commutation. $P_{a-1,b-2}$ commutes with $(T_{b-1} + 1)$. Indeed, $P_{a-1,b-2} = R'_{a-1} R'_a \cdots R'_{b-2}$ is a product of factors $(T_j + 1)$ for $j \leq b - 3$ (the maximum index of a T -factor inside R'_{b-2} is $b - 3$). Each such T_j has $|j - (b - 1)| \geq 2$, so T_j commutes with T_{b-1} , hence so does the product. Therefore (2) becomes

$$P_{a,b} = (T_{a-1} + 1)(T_a + 1) \cdots (T_{b-2} + 1)(T_{b-1} + 1) P_{a-1,b-2} R'_{b-1}.$$

Since $P_{a-1,b-2} R'_{b-1} = P_{a-1,b-1}$ by definition, we get (1) for $P_{a,b}$. $\square$

Remark 3. Specialising to $b = a + 1$ recovers the commutation identity used in the May-13 night same-row-dies paper for $(3, 3, 1^q)$:

$$R'_a R'_{a+1} = (T_{a-1} + 1)(T_a + 1) R'_{a-1} R'_a.$$

The Shift Lemma is the natural generalisation to chains of arbitrary length.

Remark 4. The Shift Lemma (1) holds as a literal operator identity inside $H_q(S_n)$ (independent of any representation), verified computationally at $P = 100\,003$, $Q = 11$ in: `~/projects/scratch/2026-05-13-shift` for $\lambda \in \{(3, 2), (2, 2, 1), (3, 1^3), (4, 1^3), (2, 2, 1^2), (3, 2, 1^2)\}$ at $n \in \{5, 6, 7\}$ and all valid (a, b) pairs.

3 Hook structure and prior ingredients

Fix $\lambda = (k, 1^{n-k})$ with $k \geq 2$ and $n \geq 2k$. We have $\ell(\lambda) = n - k + 1$ and $j_0(\lambda) = n - k + 2$. The removable corners of λ are:

$$c_1 := (1, k) \quad (\text{length-preserving}), \quad c_* := (n - k + 1, 1) \quad (\text{column-1 bottom}).$$

3.1 Phase A and the E_+ decomposition

Theorem 5 (Phase A endpoint, May-19). *For every $H_q(S_n)$ -module V and every $1 \leq j \leq n - 1$, $(T_j + 1)(T_{j-1} + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

Lemma 6 (Decomposition of E_+ for hooks). *For $\lambda = (k, 1^{n-k})$ with $k \geq 3$,*

$$E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S, \quad (3)$$

where

- $\mu_1 := \lambda \setminus \{c_1, c_*\} = (k - 1, 1^{n-k-1})$ on $n - 2$ letters;
- $\Phi_* : V_{\mu_1} \hookrightarrow V_\lambda$ is the $+q$ -eigenvector embedding of the T_{n-1} -2-block at the corner pair $\{c_1, c_*\}$ (May-13 r1 Defn 2.1);
- S is the span of seminormal basis vectors v_T for SYTs of λ with the letters $\{n - 1, n\}$ at the same-row pair $\{(1, k - 1), (1, k)\}$ at row 1.

For $k = 2$, $S = 0$ and $E_+ = \Phi_*(V_{\mu_1})$ already.

Proof. T_{n-1} acts on v_T by the Hoefsmit rule depending on the relative positions of $n - 1, n$ in T . Eigenvalue $+q$ arises in two cases:

- (1) $\{n - 1, n\}$ at a corner pair (one length-pres, one column-1 bottom) in different row and column: T_{n-1} is a 2×2 Hoefsmit block, one-dim $+q$ -eigenvector inside. The only such pair for a hook is $\{c_1, c_*\}$. The $+q$ -eigenvectors are exactly the $\Phi_*(v_{T'})$ for $T' \in \text{SYT}(\mu_1)$ (May-13 r1 Defn 2.1).
- (2) $\{n - 1, n\}$ at a same-row pair $\{(i, \lambda_i - 1), (i, \lambda_i)\}$: $T_{n-1}v_T = qv_T$.

For a hook $\lambda = (k, 1^{n-k})$, the same-row pair is valid at row i iff $\lambda_i \geq 2$ and $(\lambda_{i+1} \leq \lambda_i - 2$ or $i = \ell)$. Only row 1 satisfies $\lambda_1 = k \geq 2$, and the condition $\lambda_2 = 1 \leq k - 2$ holds iff $k \geq 3$. Hence for $k \geq 3$ there is a unique same-row pair at row 1, $\{(1, k - 1), (1, k)\}$, and the corresponding S is the span of seminormal vectors with $n - 1, n$ at this pair. The remaining cells of λ form the sub-partition $\nu' := \lambda \setminus \{(1, k - 1), (1, k)\} = (k - 2, 1^{n-k})$ on $n - 2$ letters.

The direct sum (3) is by construction (distinct seminormal-basis supports), and orthogonal to the seminormal pairing. $\square$

3.2 The Φ_* -commutation

Lemma 7 (Φ_* commutation, May-13 r1 Lemma 3.1). *For any partition λ with a single length-pres corner c_1 and column-1 bottom c_* (hypotheses (H1)–(H3) of the May-13 r1 paper),*

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} |_{\Phi_*(V_{\mu_1})} = (T_\ell + 1)(T_{\ell+1} + 1) \cdots (T_{n-2} + 1) \Phi_*(R'_\ell^{(\mu_1)} R'_{\ell+1}^{(\mu_1)} \cdots R'_{n-2}^{(\mu_1)}).$$

In particular,

$$R'_{\ell+1} \cdots R'_{n-1} \Phi_*(V_{\mu_1}) = (T_\ell + 1) \cdots (T_{n-2} + 1) \Phi_*(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1}).$$

Proof. See May-13 r1 paper, Section 3 and the May-15 paper. The proof is purely operator-theoretic: Φ_* is $H_q(S_{n-2})$ -equivariant under $\langle T_1, \dots, T_{n-3} \rangle$, and the iteration uses only $(T_j + 1)$ factors with $j \leq n - 2$, all of which (after the initial $(T_{n-2} + 1)$) commute through Φ_* via index-gap arguments. Hooks satisfy (H1)–(H3): one length-pres corner $c_1 = (1, k)$, column-1 bottom $c_* = (n - k + 1, 1)$, axial distance $d = -(n - k) - (k - 1) = -(n - 1)$, $|d| = n - 1 \geq 2$ for $n \geq 3$. $\square$

3.3 The same-row iso Ψ

Lemma 8 (Same-row iso). *For $\lambda = (k, 1^{n-k})$ with $k \geq 3$, the map*

$$\Psi : S \xrightarrow{\sim} V_{\nu'}, \quad \nu' = (k - 2, 1^{n-k}) \text{ on } n - 2 \text{ letters,}$$

sending $v_T \in S$ (where T has $n - 1, n$ at $(1, k - 1), (1, k)$) to $v_{T|_{\nu'}}$ (the restriction of T to the letters $\{1, \dots, n - 2\}$ filling the cells of ν') is a $\mathbb{Q}(q)$ -linear isomorphism, $H_q(S_{n-2})$ -equivariant under $\langle T_1, \dots, T_{n-3} \rangle$.

Proof. For $T \in S$, the letters $n - 1, n$ are at $(1, k - 1), (1, k)$, which are NOT cells of ν' . The letters $\{1, \dots, n - 2\}$ fill the cells of ν' (namely row 1 of length $k - 2$ and the column-1 tail of length $n - k + 1$). So $T|_{\nu'}$ is a well-defined SYT of ν' . The map is bijective by construction.

For equivariance: for $j \leq n - 3$, T_j acts on letters $j, j + 1$, both $\leq n - 2$, hence in cells of ν' . The Hoefsmit action of T_j on v_T depends only on the positions of $j, j + 1$ in T , which coincide with their positions in $T|_{\nu'}$. Hence $T_j \Psi^{-1}(v) = \Psi^{-1}(T_j v)$. $\square$

Remark 9. Since Ψ is T_1 -equivariant (because $1 \leq n - 3$ for $n \geq 4$), Ψ preserves the T_1 -eigenspaces. In particular $\Psi(S \cap E_1^-) = E_1^-|_{V_{\nu'}}$. Similarly for Φ_* (May-13 r1 Lemma 2.1): Φ_* is T_1 -equivariant for $n \geq 4$, so $\Phi_*(E_1^-|_{V_{\mu_1}}) \subseteq E_1^-|_{V_\lambda}$.

4 Main theorem and proof

Theorem 10 (Hook j -formula, all k). *For every $k \geq 1$ and every $n \geq 2k$, the hook $\lambda = (k, 1^{n-k})$ satisfies*

$$B_{j_0(\lambda)}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}, \quad j_0(\lambda) = n - k + 2. \quad (4)$$

Proof. Strong induction on k .

Base case $k = 1$: $\lambda = (1^n)$ is the sign representation; T_i acts as -1 for every i . So $R'_j V_\lambda = 0$ for every $j \geq 2$ (each factor $(T_i + 1) = 0$ on V_λ). Hence $B_j^{(\lambda)} V_\lambda = 0$ trivially $\subseteq E_1^-|_{V_\lambda}$, for every j .

Base case $k = 2$: $\lambda = (2, 1^{n-2})$, May-11 paper (2026-05-11-j-formula.tex, Theorem on $\lambda = (2, 1^{n-2})$).

Inductive step $k \geq 3$: Assume (4) for all $k' < k$ (in their rank-zero regime). Let $\mu_1 = (k - 1, 1^{n-k-1})$ and $\nu' = (k - 2, 1^{n-k})$, both on $n - 2$ letters. Both are hooks; rank-zero regime for μ_1 is $n - 2 \geq 2(k - 1)$, i.e., $n \geq 2k$ (satisfied); similarly ν' at $n - 2 \geq 2(k - 2) = 2k - 4$, $n \geq 2k - 2$ (weaker; satisfied). By IH:

$$B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} \subseteq E_1^-|_{V_{\mu_1}}, \quad B_{j_0(\nu')}^{(\nu')} V_{\nu'} \subseteq E_1^-|_{V_{\nu'}}. \quad (5)$$

By Phase A (Theorem 5) and the E_+ decomposition (Lemma 6),

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S.$$

Hence

$$B_{j_0(\lambda)}^{(\lambda)} V_\lambda = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} R'_n V_\lambda = R'_{j_0} \cdots R'_{n-1} (\Phi_*(V_{\mu_1}) + S).$$

We show each summand is in $E_1^-|_{V_\lambda}$.

Part 1: Φ_* -piece is in E_1^- .

By Lemma 7 applied to λ (hook, $r = 1$):

$$R'_{j_0} \cdots R'_{n-1} \Phi_*(V_{\mu_1}) = (T_\ell + 1)(T_{\ell+1} + 1) \cdots (T_{n-2} + 1) \Phi_* \left(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} \right),$$

where $\ell = n - k + 1$ and $j_0(\mu_1) = (n - 2) - (k - 1) + 2 = n - k + 1$. By the IH on μ_1 (first part of (5)), $B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} \subseteq E_1^-|_{V_{\mu_1}}$. By Remark 9, Φ_* preserves E_1^- , so $\Phi_*(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1}) \subseteq E_1^-|_{V_\lambda}$. The outer T -factors $(T_\ell + 1), (T_{\ell+1} + 1), \dots, (T_{n-2} + 1)$ all have index $\geq \ell = n - k + 1 \geq 3$ (since $n \geq 2k \geq k + 2$ for $k \geq 2$), hence commute with T_1 (index gap ≥ 2) and preserve $E_1^-|_{V_\lambda}$. Therefore

$$R'_{j_0} \cdots R'_{n-1} \Phi_*(V_{\mu_1}) \subseteq E_1^-|_{V_\lambda}.$$

Part 2: S -piece dies (same-row dies).

By the Shift Lemma (Lemma 2) with $a = j_0$ and $b = n - 1$,

$$R'_{j_0} \cdots R'_{n-1} = (T_{j_0-1} + 1)(T_{j_0} + 1) \cdots (T_{n-2} + 1) R'_{j_0-1} R'_{j_0} \cdots R'_{n-2}. \quad (6)$$

Note $j_0 - 1 = n - k + 1 = \ell$, and the right-hand inner factor is $P_{\ell, n-2}$.

Apply to S . The operators R'_j for $j \leq n - 2$ all involve only T_i with $i \leq n - 3$, so by Lemma 8 (equivariance of Ψ under $\langle T_1, \dots, T_{n-3} \rangle$),

$$P_{\ell, n-2}|_S = \Psi^{-1}(P_{\ell, n-2}^{(\nu')} V_{\nu'}) = \Psi^{-1}(B_{\ell}^{(\nu')} V_{\nu'}),$$

where the rightmost identification is just unwinding the notation: for $\nu' = (k - 2, 1^{n-k})$ on $n - 2$ letters,

$$\ell(\nu') = 1 + (n - k) = n - k + 1 = \ell, \quad j_0(\nu') = n - k + 2 = j_0(\lambda),$$

and $|\nu'| = n - 2$, so $P_{\ell, n-2}^{(\nu')} = R_{\ell}^{(\nu')} R_{\ell+1}^{(\nu')} \cdots R_{|\nu'|}^{(\nu')} = B_{\ell(\nu')}^{(\nu')} V_{\nu'}$.

This is the B -image *one step below* the sharp $j_0(\nu')$:

$$B_{\ell(\nu')}^{(\nu')} V_{\nu'} = R_{\ell(\nu')}^{(\nu')} \cdot B_{j_0(\nu')}^{(\nu')} V_{\nu'} \subseteq R_{\ell(\nu')}^{(\nu')} \cdot E_1^-|_{V_{\nu'}} = 0. \quad (7)$$

The middle inclusion is the IH on ν' (second part of (5)); the final equality is because $R_{\ell(\nu')}^{(\nu')}$ has rightmost factor $(T_1 + 1)$, which annihilates $E_1^-|_{V_{\nu'}}$.

Therefore $P_{\ell, n-2}|_S = \Psi^{-1}(0) = 0$, so by (6),

$$R'_{j_0} \cdots R'_{n-1} S = (T_{j_0-1} + 1) \cdots (T_{n-2} + 1) P_{\ell, n-2} S = 0.$$

Conclusion: Combining Parts 1 and 2, $B_{j_0(\lambda)}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda} + 0 = E_1^-|_{V_\lambda}$. This closes the induction. $\square$

5 Consequences

Corollary 11 (Rank-zero theorem for all hooks). *For every $k \geq 2$ and every $n \geq 2k$,*

$$\Pi^{S_n} |_{V_{(k, 1^{n-k})}} = 0.$$

Proof. By Theorem 10, $B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-$. Iteratively descending: $B_{j_0-1}^{(\lambda)} = R'_{j_0-1} B_{j_0}^{(\lambda)} \subseteq R'_{j_0-1} E_1^- = 0$ (rightmost $(T_1 + 1)$ kills E_1^-). So $B_j^{(\lambda)} = 0$ for every $2 \leq j \leq j_0 - 1$; in particular $\Pi^{S_n} |_{V_\lambda} = B_2^{(\lambda)} = 0$. $\square$

Corollary 12 ($\dim B_{j_0}^{(\lambda)} \leq 1$ for all hooks). *For every $k \geq 2$ and every $n \geq 2k$,*

$$\dim B_{j_0(\lambda)}^{(\lambda)} V_\lambda \leq 1.$$

Proof. Same-row dies (Part 2 of the proof of Theorem 10) gives the equality (not just inclusion)

$$B_{j_0(\lambda)}^{(\lambda)} V_\lambda = (T_\ell + 1) \cdots (T_{n-2} + 1) \Phi_*(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1})$$

(May-13 r1 Theorem 3, unconditional now that same-row-dies is proved for hooks). Since Φ_* is injective and outer T -factors don't increase dimension, $\dim B_{j_0}^{(\lambda)} \leq \dim B_{j_0(\mu_1)}^{(\mu_1)}$. By induction down to $k = 2$ where $\dim B_{j_0}^{(2, 1^{n-2})} = 1$ (May-11), the bound ≤ 1 propagates. $\square$

Remark 13 (On $\dim = 1$). The matching lower bound $\dim B_{j_0}^{(\lambda)} \geq 1$ (giving $\dim = 1$ and the Catalan-many support of the May-12 hook paper's Conjecture 5.1) is a separate question — it requires showing the outer T -factors are not annihilating on the Φ_* -image. Direct seminormal-basis tracking establishes $\dim = 1$ for $k \leq 4$ at $n \leq 9$ (May-12, May-12 evening-3); the full structural proof at general (k, n) is unaddressed here. This paper focuses on the E_1^- -containment, which is the j -formula proper.

Corollary 14 (Unconditional j -formula on $(4, 2, 1^{n-6})$). *The May-12 evening-3 paper's j -formula for $\lambda = (4, 2, 1^{n-6})$ at $n \geq 10$ is now unconditional: $B_{n-3}^{(\lambda)} V_\lambda \subseteq E_1^- |_{V_\lambda}$, and $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$.*

Proof. The May-12 evening-3 paper assumes (Theorem 3.2 of that paper) the hook j -formula at $k = 4$ for $\mu_B = (4, 1^{n-6})$ on $n - 2$ letters. By Theorem 10 (at $k = 4$), this is now established structurally. The rest of the May-12 evening-3 argument goes through verbatim, giving the upper bound ≤ 3 and (with the explicit construction) the matching lower bound. $\square$

6 Computational verification

Scripts. `~/projects/scratch/2026-05-13-shift-lemma/verify.py` (Shift Lemma + hook j -formula) and `verify_extras.py` (same-row-dies on $(k, k, 1^q)$, larger hooks). Mod $P = 100003$, $Q = 11$.

Shift Lemma. Verified as a literal operator identity in $\text{End}(V_\lambda)$ for $\lambda \in \{(3, 2), (2, 2, 1), (3, 1^3), (4, 1^3), (2, 2, 1^2), (3, 2, 1), (2, 1^4)\}$ at all valid (a, b) pairs in $2 \leq a \leq b \leq n$. All pass.

Hook j -formula. Verified for $(k, n) \in \{(2, 4), (2, 5), (2, 6), (3, 6), (3, 7), (3, 8), (4, 8), (4, 9), (4, 10), (5, 10)\}$. In every case, $(T_1 + 1)$ applied to $B_{j_0}^{(\lambda)} V_\lambda$ has rank 0, confirming the E_1^- containment.

Same-row dies for $(k, k, 1^q)$, sanity. Same-row part S has the predicted dimension $f^{(k, k-2, 1^q)}$ and the prediction $P_{j_0(\lambda), n-1} S = 0$ holds for $(3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$ (recovering the May-13 night theorem). For $(4, 4, 1^q)$, the prediction holds only when $q \geq 4$ (the regime where $\nu = (4, 2, 1^q)$ enters its own rank-zero regime); at $q = 2, 3$, $\text{rank}(P_{j_0, n-1} S) = 3$, consistent with the failure of ν 's j -formula in its boundary regime. This sharpens what "same-row dies" actually requires: the j -formula on ν , which is itself shape- and regime-sensitive.

7 Beyond hooks: the framework's reach

The proof of Theorem 10 factors into three independent ingredients:

- (1) **Phase A + E_+ decomposition:** $R'_n V_\lambda = E_+$ as a direct sum of Φ_X -pieces (multi-corner) and S_i -pieces (same-row). General fact.
- (2) **Φ -commutation:** each multi-corner piece is handled by the May-13 r1 / May-15 Φ -commutation (combined with the multi-corner reduction of May-12 evening-3).
- (3) **Shift Lemma + iso:** each same-row piece reduces to the j -formula on the sub-shape $\nu'_i = \lambda \setminus \{\text{row-}i \text{ same-row pair}\}$ via the Shift Lemma and the iso $\Psi_i : S_i \xrightarrow{\sim} V_{\nu'_i}$.

The induction on k for hooks works because both subproblems (μ_1 and ν') are hooks with smaller arm length. For general shapes, the induction descends along multi-corner sub-shapes and same-row sub-shapes; both decrease the size of the "decoration shape" $\lambda^{(\geq 2)}$ (the column-1-deleted partition).

Concrete next applications.

- **j -formula on $(k, 2, 1^q)$ for $k \geq 4$.** $r = 2$ multi-corner shape with $\mu_A = (k-1, 2, 1^{q-1})$ (recursive), $\mu_B = (k, 1^{q-1})$ (hook, this paper). The May-12 evening-3 machinery handles $k = 4$; iterating in k should work.
- **Same-row dies on $(k, k, 1^q)$ for $k = 4, q \geq 4$.** Reduces via the Shift Lemma to the j -formula on $\nu = (k, k-2, 1^q)$ in ν 's rank-zero regime. For $k = 4$, $\nu = (4, 2, 1^q)$ on $m = q + 6$ letters; rank-zero requires $q \geq 4$. In that regime, Corollary 14 supplies the needed j -formula on ν , so same-row dies for $(4, 4, 1^q)$ at $q \geq 4$ follows. Below $q = 4$, ν is at or below the rank-zero boundary and the same-row part does *not* die (verified computationally at $(4, 4, 1^2)$ and $(4, 4, 1^3)$, where $\text{rank}(P_{j_0, n-1} S) = 3 \neq 0$).
- **j -formula on $(k, k-2, 1^q)$ for $k \geq 5$.** Same-row pair on row 2 gives $\nu' = (k, k-4, 1^q)$; recursion in $k-2$. Requires also the Φ -piece argument with μ_1, μ_2 themselves $r = 2$ non-hook shapes.

The hook case (this paper) is the cleanest application of the framework; the general theorem " j -formula in maximum generality, in the stable regime" is within reach but requires combining all three ingredients carefully across nested inductions.

8 Honest status

What's proved unconditionally. Theorem 10: the j -formula $B_{j_0(\lambda)}^{(\lambda)} V_\lambda \subseteq E_1^- |_{V_\lambda}$ for every hook $\lambda = (k, 1^{n-k})$ with $k \geq 1$ and $n \geq 2k$. This subsumes May-11 ($k = 2$) and May-12 ($k = 3$) and extends to all k .

What's a corollary. Rank-zero theorem for all hooks in the rank-zero regime (Corollary 11); $\dim B_{j_0}^{(\lambda)} \leq 1$ (Corollary 12); unconditional j -formula on $(4, 2, 1^{n-6})$ (Corollary 14).

What's not proved. The matching lower bound $\dim = 1$ (May-12 paper's Conjecture 5.1, the Catalan support phenomenon). My framework gives $\dim \leq 1$ via the same-row-dies machinery, but the non-vanishing of the outer T -factor action on the Φ_* -image is not established here.

What it took. The proof is short — three pages of ingredients, two pages of induction. The key was identifying the Shift Lemma (Section 2) as the operator-theoretic mechanism that aligns the same-row iteration on λ with the sharp j -formula iteration on ν' . Once the Shift Lemma is in hand, the rest is routine induction.

The Shift Lemma is also the right framework for the more general same-row-dies thread: any shape λ with a same-row pair admits the same one-step reduction, modulo a j -formula on the sub-shape.